

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	13 July 2026
Team ID	SWTID-2026-8698
Project Name	Nutrition Assistant
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Input Validation Password Encryption (bcryptjs)
FR-2	User Login	Login using Email & Password JWT Token Generation User Authentication
FR-3	User Profile Management	Add Personal Details Update Age, Height, Weight & Goal Save Profile Information
FR-4	Nutrition Plan Generation	Calculate Daily Calories Calculate Protein, Carbohydrates & Fats Generate Personalized Nutrition Plan
FR-5	Meal Recommendations	Display Breakfast, Lunch, Dinner & Snacks Show Calorie & Macronutrient Information
FR-6	Nutrition History	Save Generated Nutrition Plans View Previous Nutrition Plans

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application should have a simple and user-friendly interface.
NFR-2	Security	User passwords should be encrypted, and authentication should use JWT.
NFR-3	Reliability	The system should provide accurate nutrition calculations and stable performance.
NFR-4	Performance	Nutrition plans should be generated quickly with minimal response time.
NFR-5	Availability	The application should be available whenever users need to access it.
NFR-6	Scalability	The system should support future features such as BMI calculation, AI recommendations, and progress tracking.